

MedTrace AI Complaint Investigation Report

Device: DEV-PULSE-OX

Framework: FDA 21 CFR Part 820

Status: completed

Complaint Summary

Severity	High
Affected component	COMP-ALARM
Symptoms	SYMPTOM_PULSE, SYMPTOM_OXIMETER, SYMPTOM_DISPLAY, SYMPTOM_FREEZES, SYMPTOM_FIRMWARE
Timeline	acute event
Raw summary	Pulse oximeter display freezes after firmware v3.4 update during alarm event queue processing. SpO2 value does not refresh for several seconds.

Root Cause Hypotheses

HYP-001: COMP-ALARM: objective evidence coverage gap

The audit scope contains 20 requirement(s) below green readiness, suggesting missing or weak objective evidence may explain the complaint.

[Source: M1 graph context and complaint state, Confidence: 0.82]

HYP-002: COMP-ALARM: traceability integrity gap

The graph shows 3 stale evidence link(s), so the complaint may originate from verification that no longer represents the current device configuration.

[Source: M1 graph context and complaint state, Confidence: 0.82]

HYP-003: COMP-ALARM: risk-control effectiveness drift

The affected area may have a risk-control weakness because 2 open CAPA record(s) exist in the device context.

[Source: M1 graph context and complaint state, Confidence: 0.82]

M4 Evidence Retrieved

Evidence	Source	Confidence	Snippet
EVID-1bf4480155bd	00_Index/DocPlus_M1_Data_Requirements.md	0.95	# DocPlus+ M1 Professional Data Requirements Prepared for: DocPlus+ M1 rebuild Device focus: Pulse Oximeter with display freeze, software update, and delayed alarm complaint Goal: Build an M1 graph that lets M2 investigate actual causes, not broad evidence...
EVID-6f1a4094a68d	06_Verification_Validation/Raw_Test_Logs/RAWLOG_Display_Task_FW-v3....	0.95	# RAWLOG_Display_Task_FW-v3.4.log regenerated from display/UI complaint rows 2026-03-01 08:00:00Z DISP CPL-100-011 OLED/LCD screen remains blank. root=OLED boost converter enable trace fractured at the flex tail bend radius; test=TC-DISP-011; requirement=RE...

EVID-5f598b0006bf	06_Verification_Validation/Test_Reports/TEST_REPORT_Complaint100_Re...	0.95	DocPlus+ Pulse Oximeter - Verification and Regression Report Page 1 Verification and Validation Package DocPlus+ Pulse Oximeter Complaint100 Regression - Firmware v3.4 Family Document ID VVR-C100-FW34 Report Type Requirements-based verification and regressi...
EVID-6f1a4094a68d	06_Verification_Validation/Raw_Test_Logs/RAWLOG_Display_Task_FW-v3....	0.95	# RAWLOG_Display_Task_FW-v3.4.log regenerated from display/UI complaint rows 2026-03-01 08:00:00Z DISP CPL-100-011 OLED/LCD screen remains blank. root=OLED boost converter enable trace fractured at the flex tail bend radius; test=TC-DISP-011; requirement=RE...
EVID-1bf4480155bd	00_Index/DocPlus_M1_Data_Requirements.md	0.87	# DocPlus+ M1 Professional Data Requirements Prepared for: DocPlus+ M1 rebuild Device focus: Pulse Oximeter with display freeze, software update, and delayed alarm complaint Goal: Build an M1 graph that lets M2 investigate actual causes, not broad evidence...
EVID-83894c0f0c71	06_Verification_Validation/Raw_Test_Logs/RAWLOG_Alarm_Event_Queue_F...	0.87	# RAWLOG_Alarm_Event_Queue_FW-v3.4.log regenerated from alarm complaint rows 2026-03-01 08:00:00Z ALARM CPL-100-051 High SpO2 alarm does not trigger. status=Closed - field correction initiated; test=TC-ALRM-051; requirement=REQ-ALRM-051; risk=RISK-ALRM-051....

Risk Assessment

Severity	4
Probability	4
RPN	16
Risk level	High
Reportable	True
Rationale	Severity came from complaint classification; probability used graph signals: stale evidence=3, open CAPAs=2, similar incidents=0, supporting evidence=6.
Citation	[Source: M1 graph readiness, complaint history, and evidence retrieval, Confidence: 0.92]

CAPA Draft

Problem Statement

Complaint intake for DocPlus+ Pulse Oximeter indicates High severity behavior affecting COMP-ALARM. [Source: DocPlus M1 Data Requirements, Confidence: 0.95]

Root Cause Hypothesis

The audit scope contains 20 requirement(s) below green readiness, suggesting missing or weak objective evidence may explain the complaint. [Source: M1 graph context and complaint state, Confidence: 0.82]

Risk Assessment

Calculated RPN is 16 under FDA 21 CFR Part 820; reportability is True. [Source: M1 graph readiness, complaint history, and evidence retrieval, Confidence: 0.92]

Corrective Action

Resolve the graph-confirmed evidence gaps for COMP-ALARM; current context shows 3 stale evidence link(s) and 2 open CAPA record(s). [Source: M1 traceability matrix and device context, Confidence: 0.90]

Preventive Action

Run Trace Decay on future changes to COMP-ALARM and any connected requirements before release approval. [Source: M2 Trace Decay workflow, Confidence: 0.88]

Effectiveness Verification

CAPA can close only when AuditShadow reports no critical or major findings for the affected DocPlus+ Pulse Oximeter requirement set. [Source: AuditShadow deterministic findings, Confidence: 0.90]

AI Reasoning Summary

Complaint: {'executive_summary': 'Complaint for DEV-PULSE-OX was classified as High severity affecting COMP-ALARM. Current deterministic RPN is 16.', 'root_cause_reasoning': ['The audit scope contains 20 requirement(s) below green readiness, suggesting missing or weak objective evidence may explain the complaint.', 'The workflow collected 6 evidence item(s) and 3 hypothesis item(s).', 'Open CAPA, stale evidence, similar incidents, and readiness gaps are the main graph-grounded drivers.'], 'capa_quality_notes': ['CAPA draft is grounded in M1 graph evidence and M2 risk output.', 'Verify effectiveness by re-running AuditShadow after corrective action updates.', 'Do not close CAPA until traceability gaps and stale evidence are resolved or justified.'], 'member_handoff': {'M1': 'Keep the knowledge grap...